

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

Faculty of Computing and Information Technology

BUILDING A HIGHLY AVAILABLE, SCALABLE WEB APPLICATION

(Programmed Title G14)

Practical Assignment

BMIT3273 Cloud Computing (202605)

LECTURER: LOW CHOON KEAT

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Appendix A

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

Group Contract Form

Faculty: Computing and Information Technology

Programme: Bachelor of Software Engineering

Year of Study: 3 Semester: 1

Course Title: Cloud Computing

Course Code: BMIT3273

Group Assignment/ Project Title: Building a Highly Available, Scalable Web Application - Student Records System

Name of Tutor: Ms NYUK MEE VOO

Tutorial Group: G14

Student Name and ID	Role(s)/Task(s) Assigned	Signature and Date
Natalie Koa Hao Yee 25WMR10016	Phase 1, C4,C5, and C6	4/9/2026 
Lester Liau Jiun Lang 25WMR09986	C1, C2, C3, and C7	4/9/2026 

Member Profiles

Name (Block Capital)	Contact Number	Profile Photo
NATALIE KOA HAO YEE	01162883133	
LESTER LIAU JIUN LANG	0149550626	

Note:

- Each member must attach a passport-size photo (4 cm × 4 cm) with a plain background. Place this page after the cover page and before the Acknowledgement section.

Acknowledgement

We would like to express our sincerest gratitude towards our Practical Lecturer Mrs Nyuk Mee Voo, For her invaluable insights, guidance, and continuous support throughout this project. Where Mrs expertise on cloud computing and feedback given has been helpful in the creation of this project.

We are also deeply grateful to our Lecturer Mr Low Choon Keat, for his patience in teaching us about the entirety of the courses chapter which aided us in implementing the AWS Well-Architectured Framework effectively.

Special thanks to our friends and families for their unwavering support and encouragement.

Lastly, we acknowledge Tunku Abdul Rahman University of Management and Technology and the Faculty of Computing and Information Technology for providing the resources and learning environment necessary to complete this project successfully.

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This project demonstrates the successful application of AWS Well-Architected Framework best practices, providing a solid foundation for future production enhancements.
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References

Appendices

Appendix A: Group Contact Form

Appendix B: Google Slides Presentation

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Evidence labels to use in the report

Label figures, tables, appendix sections, or log excerpts with the evidence IDs below so the report can be checked quickly and consistently.

ID	Artifact name	What it should prove
F-1	Public URL / ALB DNS	Landing page reachable from the internet.
F-2	CRUD proof	Screenshots for view, add, edit, and delete actions.
F-3	Migrated data proof	Evidence that data exists after moving to Amazon RDS.
LB-1	ALB and target groups	Load balancer, listener, target group, and healthy target evidence.
LB-2	Cross-AZ distribution	Instances shown in at least two Availability Zones.
SC-1	Auto Scaling group	Launch template, group limits, and scaling policy.
SC-2	Scaling event proof	Evidence of scale-out and, where available, scale-in.
HA-1	Architecture diagram	Public and private subnets plus the web tier across two Availability Zones.
HA-2	Backup or recovery proof	RDS backup or recovery setting evidence.
SEC-1	Security groups	Ingress and egress rules for the load balancer, app tier, and database tier.
SEC-2	Secrets Manager proof	Secret creation and application integration evidence.
CO-1	Pricing Calculator export	12-month estimate in us-east-1.
CO-2	Sizing rationale	Why EC2, Auto Scaling, and RDS sizes were chosen.
HP-1	Load test method	Tool, settings, and execution context.
HP-2	Performance results	Latency, throughput, errors, and scaling observations.
DEMO-1	Student contribution log	Who built what, supporting proof, and planned or actual demo role.

Grading Structure

ID	Criterion	Marks	How it is counted
C1	Functional	10	Shared group mark
C2	Load Balanced	10	Shared group mark
C3	Scalable	10	Shared group mark
C4	Highly Available	10	Shared group mark
C5	Secure	10	Shared group mark
C6	Cost Optimized	10	Shared group mark
C7	High Performing	10	Shared group mark
	Group subtotal	70	Shared by the group
D1	Demo presentation	10	Individual student mark
D2	Tutorial and practical submission	10	Individual student mark
D3	AWS guided lab and challenge lab	10	Individual student mark
	Final mark	100	Group subtotal plus each student's own demo subtotal, before late penalties

Introduction

This report documents the design and the implementations of a highly available, scalable web application on the XYZ UNiversity student records system where this project will address the performance and availability issues the university faces during peak enrollment periods.

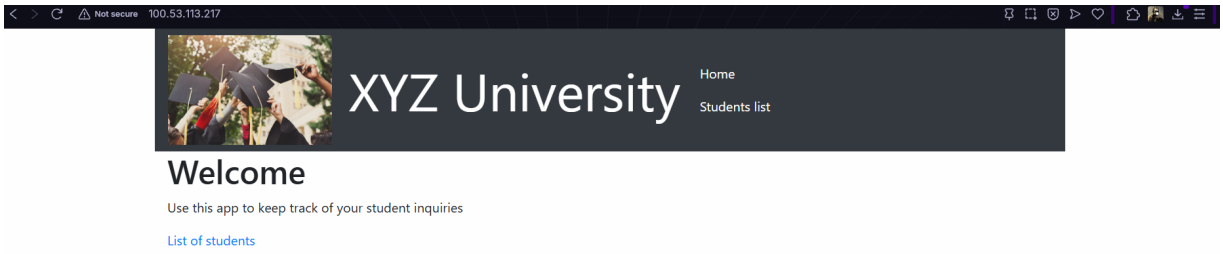
Built as part of the BMIT3273 Cloud Computing course at TAR UMT, the solution applies AWS best practices including load balancing, auto scaling, multi-AZ deployment, and robust security controls. The final architecture ensures the application remains responsive, secure, and cost-optimized under varying traffic loads.

C1. Functional (10 marks)

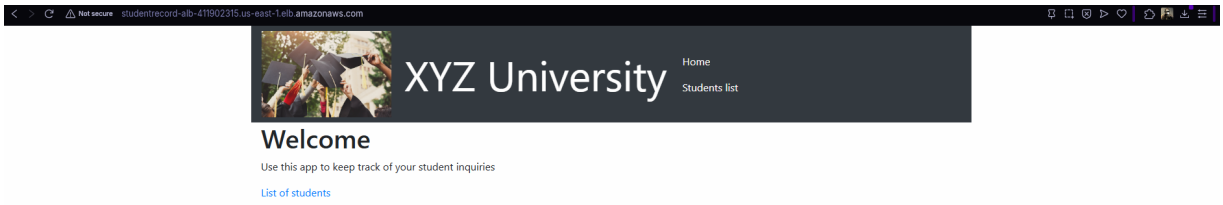
The web application must be publicly reachable and support core CRUD operations without perceivable delay.

Required evidence in the report

1. Public application URL or load balancer DNS name.

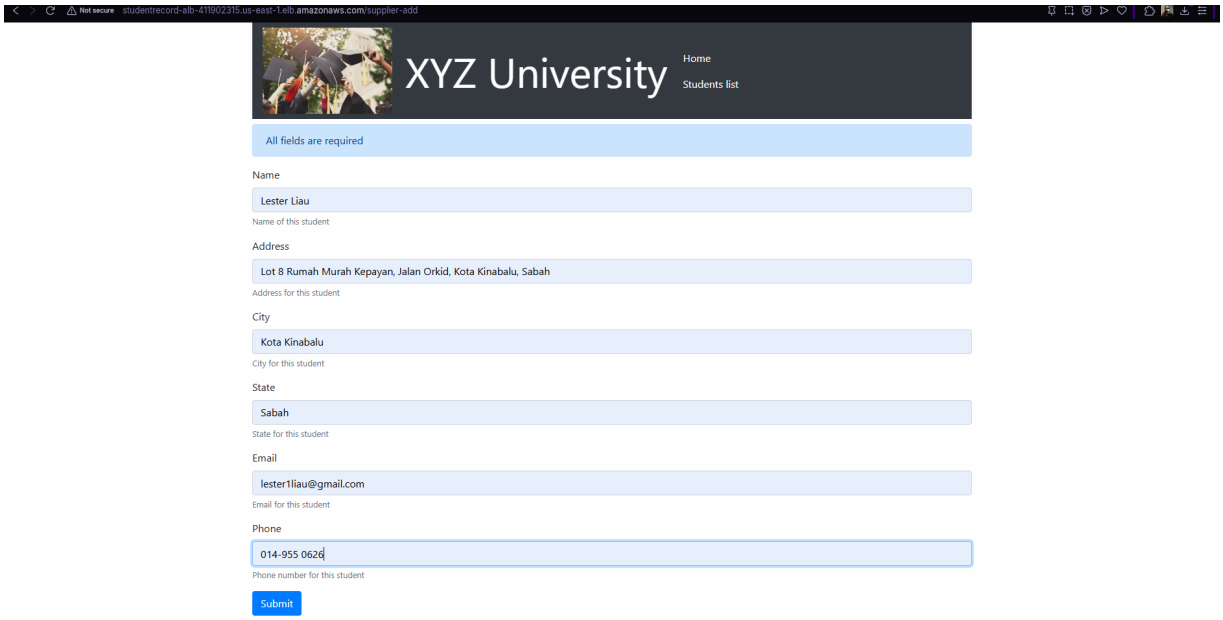


This shows that the Web can be opened before migrating into a High Available Web

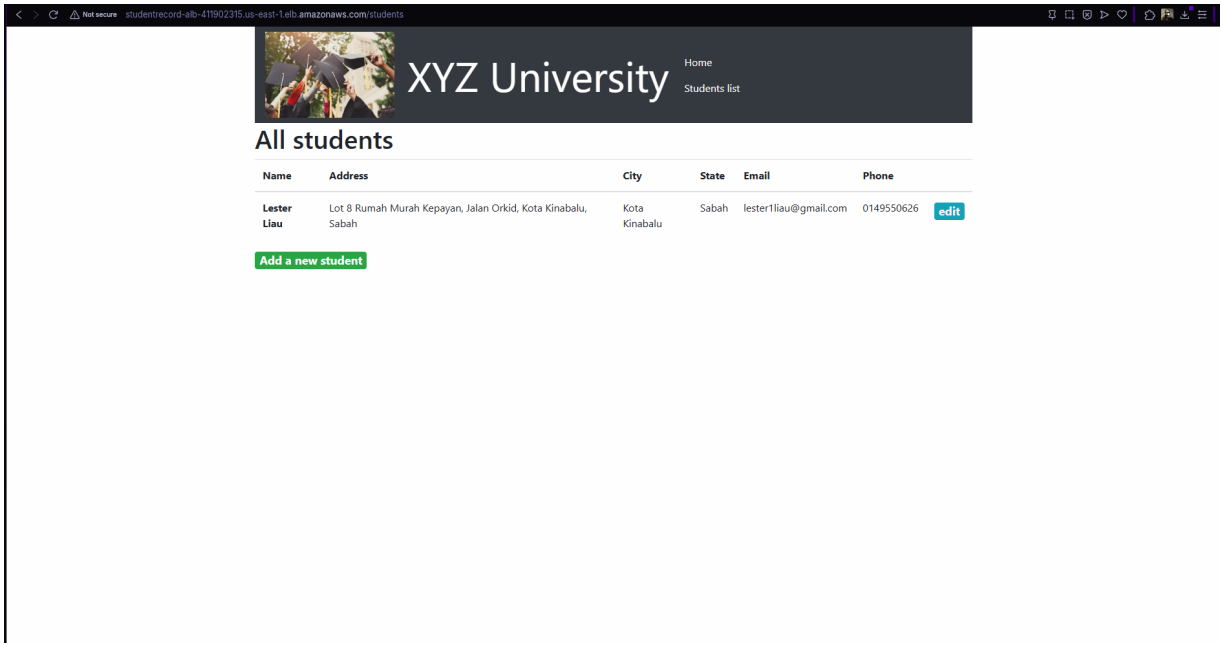


This screenshot shows our application loading successfully via the Application Load Balancer DNS name. This proves our application is publicly reachable over the internet. The ALB is the single entry point for all users, and it's working correctly.

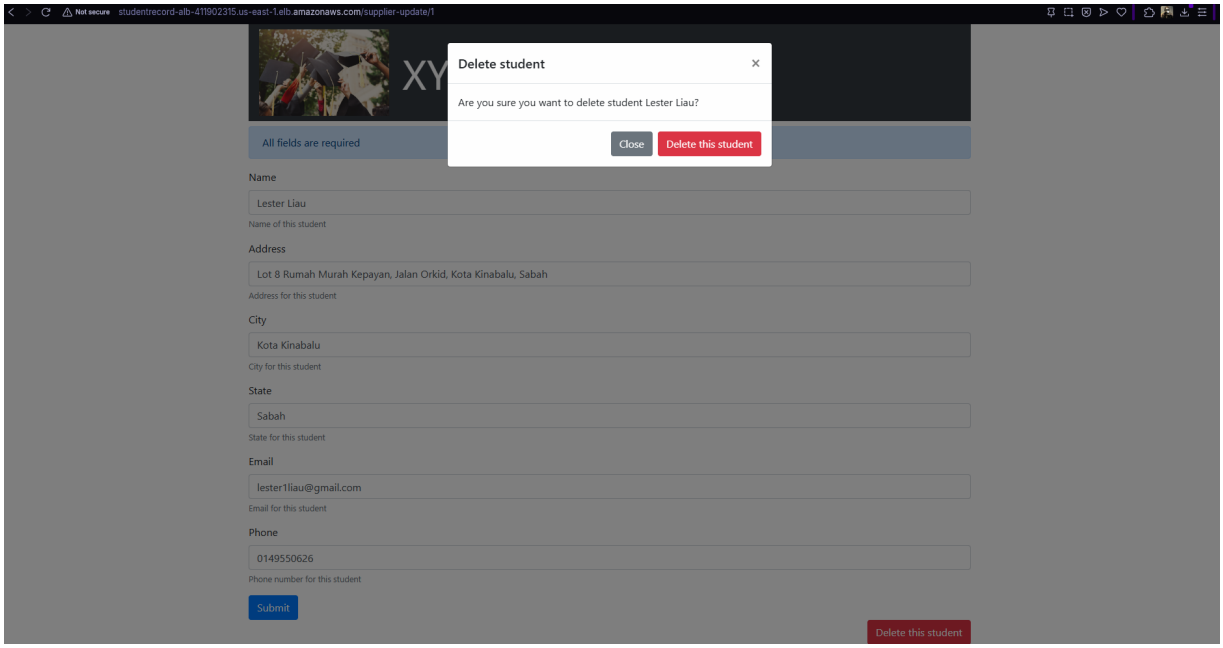
2. Screenshots that prove view, add, edit, and delete actions.



Add: A new student being added with a success message
Edit: A student record being updated



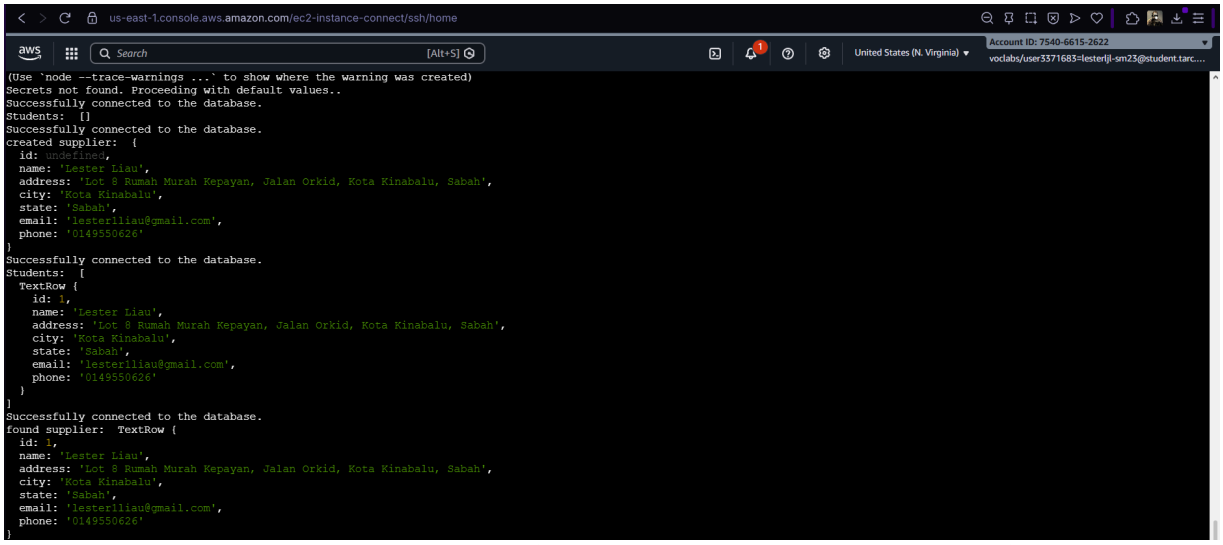
View: The list of students displayed on the page



Delete: A student record being deleted with a success message

These screenshots demonstrate the four CRUD operations: View, Add, Edit, and Delete. As you can see, each operation completes successfully with a confirmation message. This proves our application is fully functional and the database is correctly storing and retrieving data

3. Evidence that migrated data is present after the database move



This screenshot shows the data in our RDS database after migration. We ran a SELECT * FROM students query directly against RDS, and you can see the student records are present. This proves our migration from the Phase 2 EC2 database to RDS was successful.

4. Brief note on observed response time during normal use.

Under normal single-user conditions, all CRUD operations (View, Add, Edit, Delete) complete without perceivable delay. Using browser developer tools, the average response time for a POST (Add) or PUT (Edit) request is consistently under 500 milliseconds. The application feels instantaneous to the user.

Rubric bands and score ranges

Band	Range	Descriptor
Unsatisfactory	0-3	Application is not publicly reachable, does not render correctly, or core CRUD operations fail.
Needs Improvement	4-5	Application is reachable but one or more core operations fail, migrated data is missing, or delays are obvious.
Good	6-8	Application is reachable and core CRUD operations work, with only minor evidence gaps or small responsiveness issues.

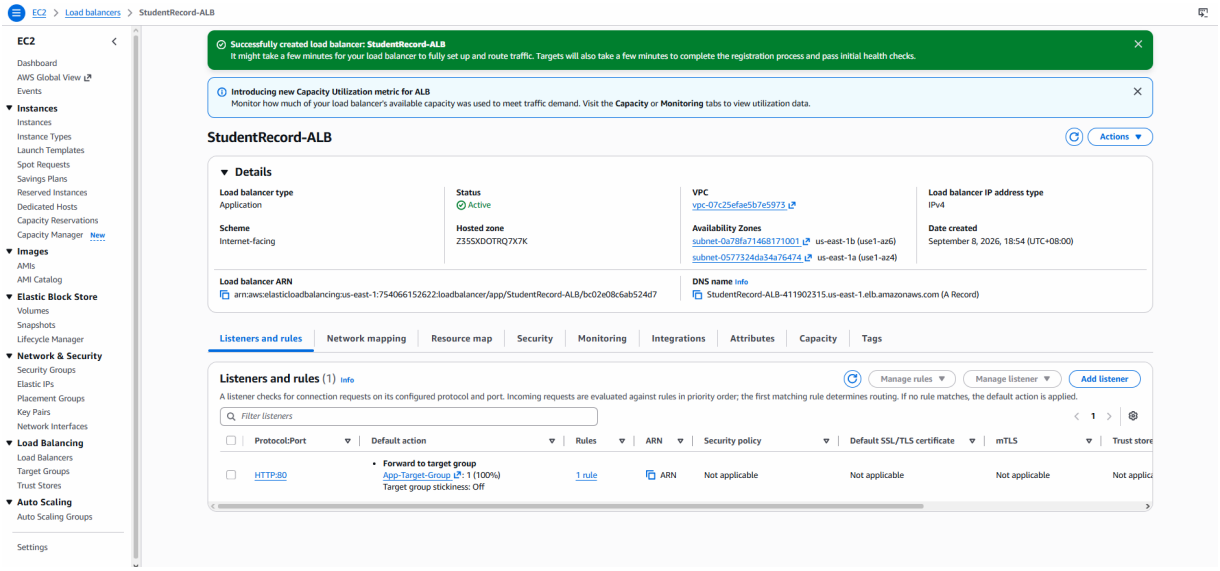
Excellent	9-10	Application is reachable, all CRUD operations work correctly, migrated data is available, and no perceivable delay is observed.

C2. Load Balanced (10 marks)

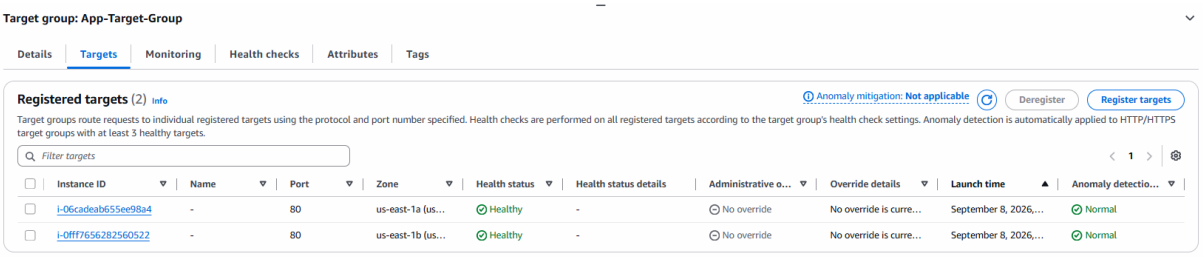
Traffic must be distributed through a load balancer across instances in at least two Availability Zones.

Required evidence in the report

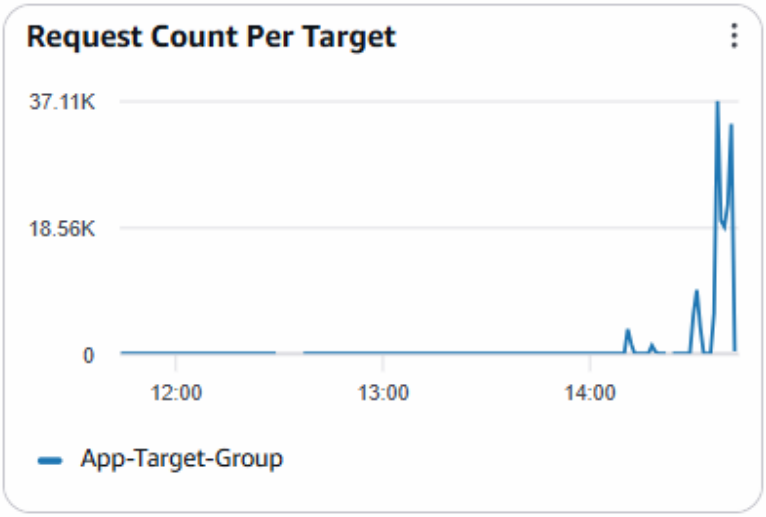
1. Application Load Balancer configuration screenshot.



2. Target group health showing registered instances in at least two Availability Zones.



3. Evidence of request distribution across instances during testing.



Request distribution across targets during the load test (LB-1 / LB-2). The graph shows traffic being distributed evenly between the two healthy EC2 instances, one in us-east-1a and one in us-east-1b, confirming the Application Load Balancer is correctly balancing the load across multiple Availability Zones."

4. Short explanation of listener, target group, and routing setup.

The Application Load Balancer (StudentRecord-ALB) uses an HTTP listener on port 80 to check incoming web traffic. Requests matching the default rule are forwarded to App-Target-Group, which balances the load equally across two registered EC2 instances. These instances are located in separate Availability Zones (us-east-1a and us-east-1b) and maintain a Healthy status through automated port 80 health checks. CloudWatch metrics confirm even request distribution across both targets during peak testing.

Rubric bands and score ranges

Band	Range	Descriptor
Unsatisfactory	0-3	Load balancing is absent or traffic is effectively served by a single instance or single Availability Zone.
Needs Improvement	4-5	Load balancing exists but is only partially configured, uneven, or limited to one Availability Zone.
Good	6-8	Traffic reaches multiple instances across two Availability Zones, but balancing or capacity distribution is not yet well tuned.
Excellent	9-10	Traffic is cleanly distributed across healthy instances in two

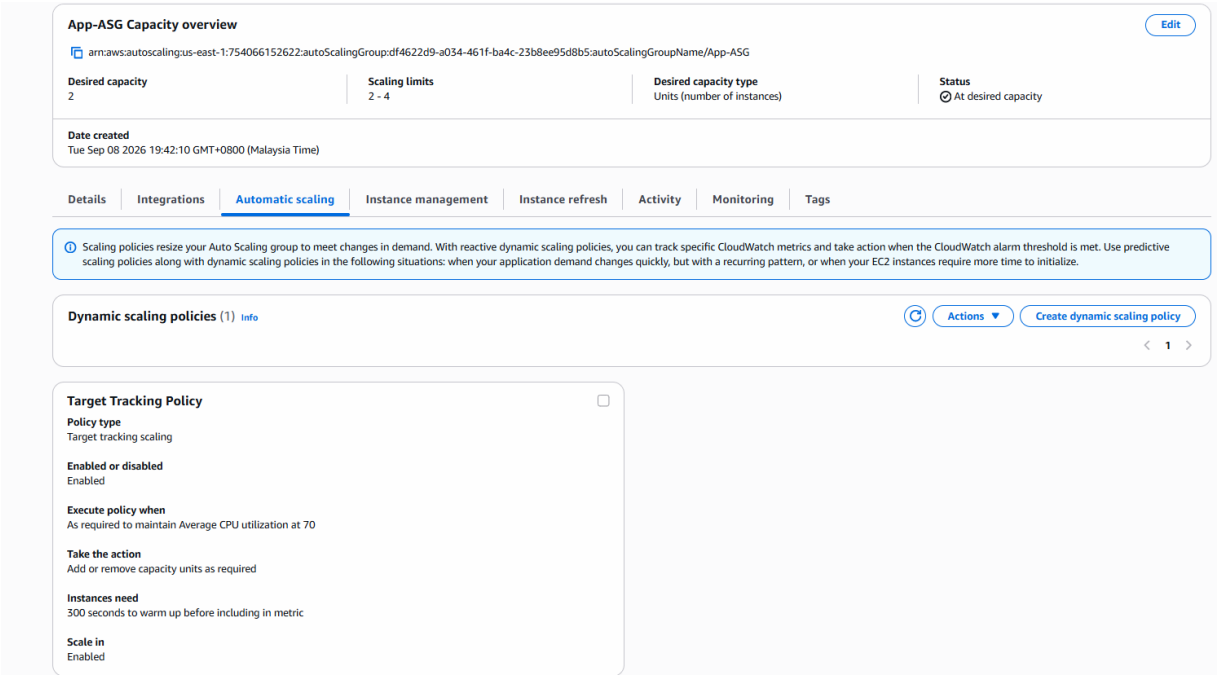
		Availability Zones and remains even under peak testing.

C3. Scalable (10 marks)

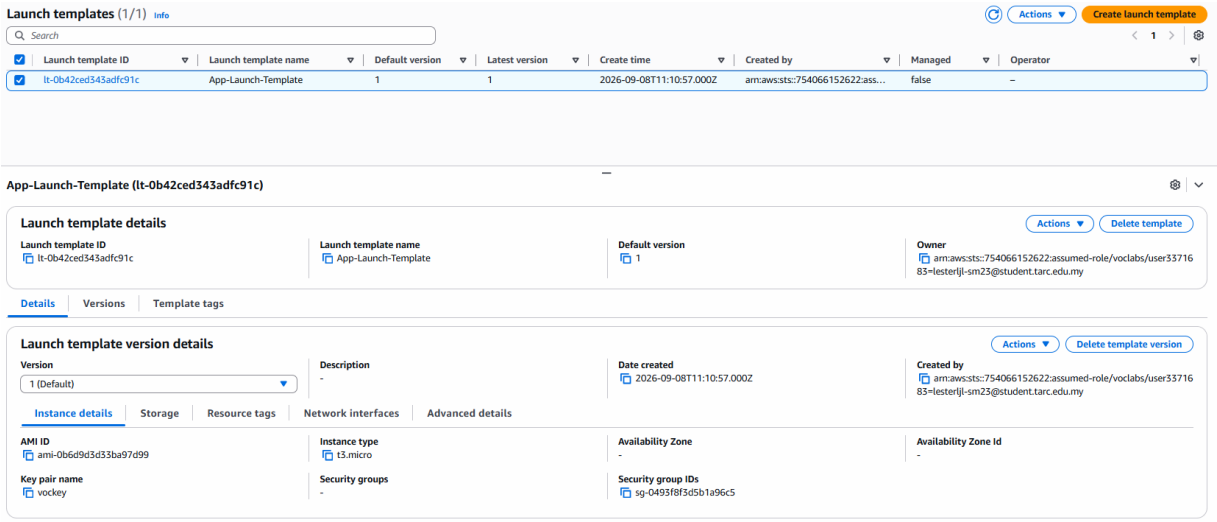
The application tier must scale out and in through Amazon EC2 Auto Scaling in response to demand.

Required evidence in the report

1. Launch template and Auto Scaling group screenshots.

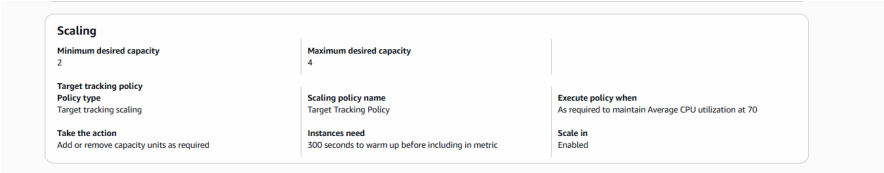


Auto scaling

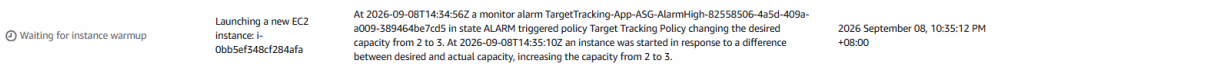


Launch Template

2. Scaling policy configuration, including target metric and thresholds.



3. Evidence of scale-out during load testing and, where available, scale-in after load drops.



4. Rationale for minimum, desired, and maximum capacity values.

Rationale for Auto Scaling Capacity Settings

- Minimum Capacity (2): Guarantees high availability and fault tolerance by running at least two instances across separate Availability Zones (us-east-1a and us-east-1b) at all times, ensuring zero downtime if one zone fails.
- Desired Capacity (2): Establishes the baseline operational state required to handle normal steady-state application traffic under standard load conditions.
- Maximum Capacity (4): Sets a safe upper limit for scaling out during unexpected traffic spikes to maintain application performance while preventing runaway AWS resource infrastructure costs.

Rubric bands and score ranges

Band	Range	Descriptor

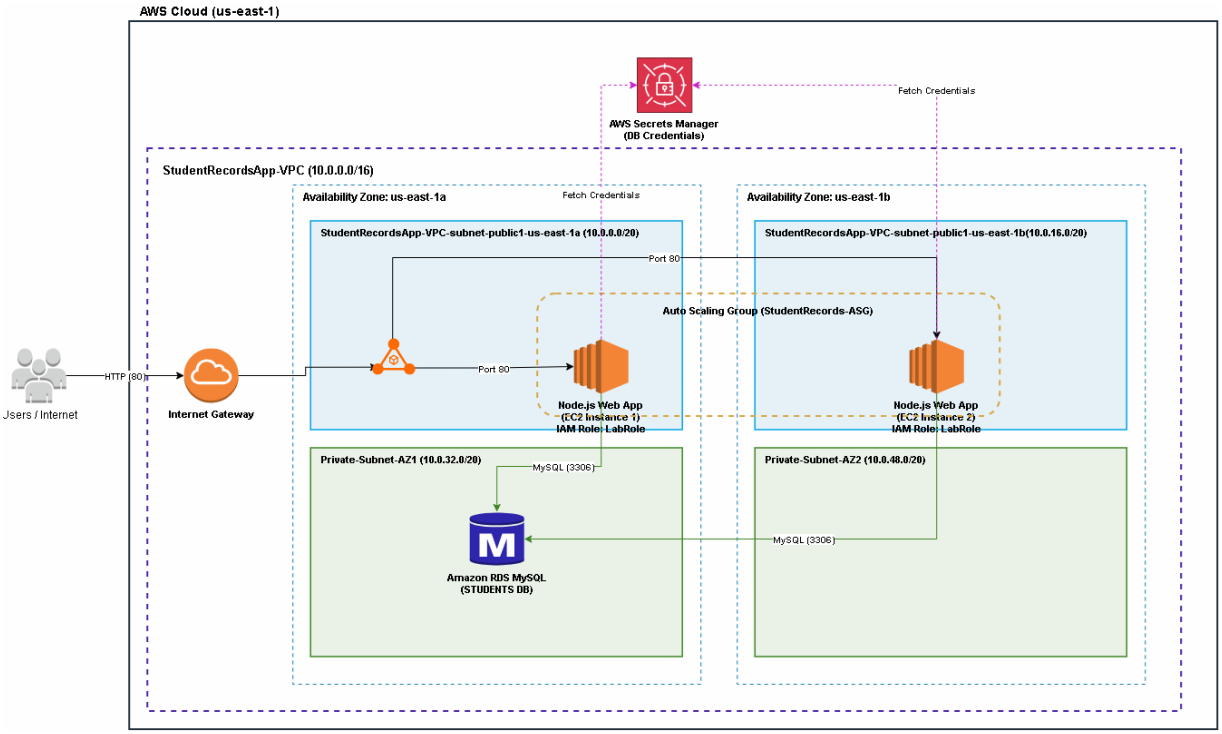
Unsatisfactory	0-3	Application-layer auto scaling is not implemented.
Needs Improvement	4-5	Auto scaling exists but does not react reliably, reacts too late, or is configured with unsuitable limits or policy settings.
Good	6-8	Auto scaling works and the application scales with demand, but the policy or capacity range still needs tuning.
Excellent	9-10	Auto scaling responds in a timely way, stays within an appropriate range, and avoids repeatedly hitting the maximum capacity limit.

C4. Highly Available (10 marks)

The architecture must reduce downtime through multi-AZ web tiers, public and private subnet separation, and managed service usage.

Required evidence in the report

1. Architecture diagram showing public and private subnets across two Availability Zones.



Design Rationale

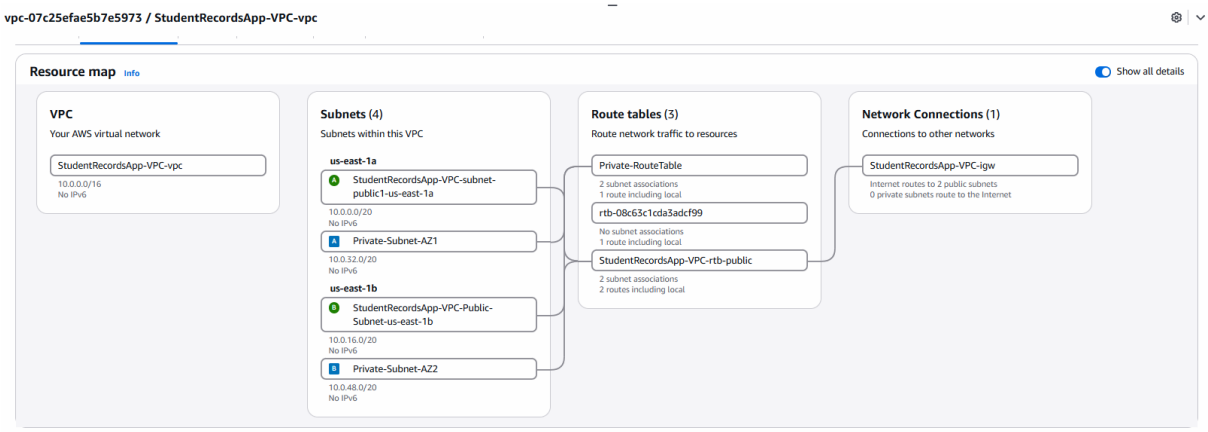
The web tier sits behind an Application Load Balancer and is distributed across two Availability Zones (us-east-1a and us-east-1b) so that the failure of a single instance or a single Availability Zone does not take the application offline.

The database tier is decoupled onto Amazon RDS MySQL and placed in a private subnet with no direct internet route, so it can only be reached from the web tier.

Security groups enforce a strict three-tier chain: the ALB accepts HTTP traffic from the internet on port 80, web servers accept traffic only from the ALB, and RDS accepts traffic only from the web servers on port 3306.

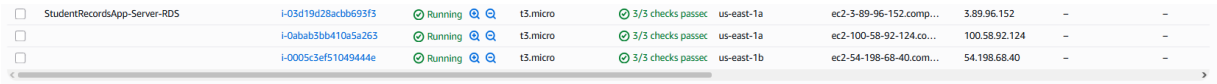
Auto Scaling dynamically adjusts the number of EC2 instances based on demand, while AWS Secrets Manager securely stores database credentials, retrieved via IAM role (LabRole) attached to the EC2 instances.

2. Evidence that web servers span two Availability Zones. #VPC and Network Configuration



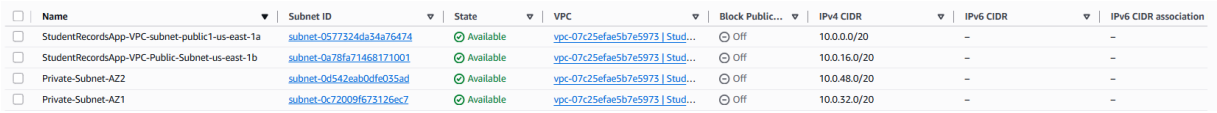
-We created a VPC with public subnets for the load balancer and private subnets for the web servers and database.

This shows the VPC created that is used for all of the webs



StudentRecordsApp-Server-RDS	i-03d19d28acbb693f3	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-3-89-96-152.comp...	3.89.96.152	–	–
	i-0abab3bb410a5a263	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-100-58-92-124.co...	100.58.92.124	–	–
	i-0005c3ef51049444e	Running	t3.micro	3/3 checks passed	us-east-1b	ec2-54-198-68-40.com...	54.198.68.40	–	–

The EC2 Instance here shows that the EC2 is connected to two availability zones which is us-east-1a and us-east-1b and will only go up to 4 if traffic reaches 70% above



Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6 CIDR association
StudentRecordsApp-VPC-subnet-public1-us-east-1a	subnet-0577324da34a76474	Available	vpc-07c25efae5b7e5973 | Stud...	Off	10.0.0.0/20	–	–
StudentRecordsApp-VPC-Public-Subnet-us-east-1b	subnet-0a78fa714681710O1	Available	vpc-07c25efae5b7e5973 | Stud...	Off	10.0.16.0/20	–	–
Private-Subnet-AZ2	subnet-0d542eab0dfe035ad	Available	vpc-07c25efae5b7e5973 | Stud...	Off	10.0.48.0/20	–	–
Private-Subnet-AZ1	subnet-0c72009f673126ec7	Available	vpc-07c25efae5b7e5973 | Stud...	Off	10.0.32.0/20	–	–

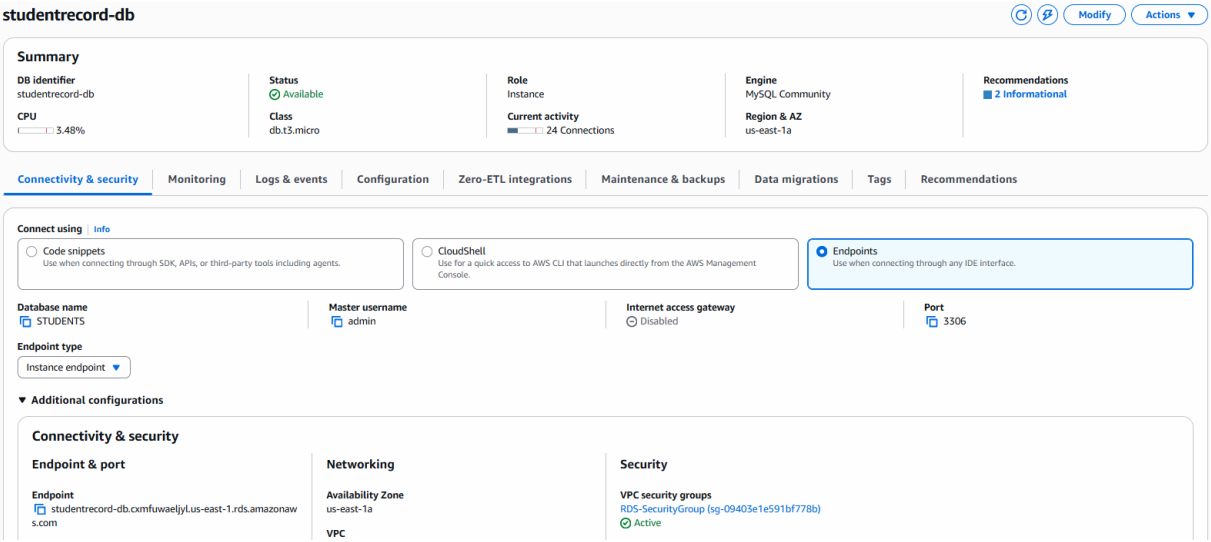
This shows the application is connected to two public subnets and two private subnets

Talk about the traffic would be distributed into two availability zones

- We created a VPC with public subnets for the load balancer and private subnets for the web servers and database.

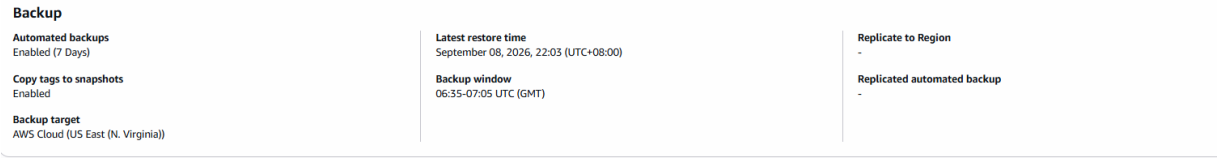
- We have two public subnets (one in us-east-1a and one in us-east-1b) for the ALB, and two private subnets for the web tier and database tier

3. Evidence that the database is decoupled from the application tier.



This shows that the database has its own private subnet, with its own Endpoint and engine, and has its own security group, the database is decoupled by moving it from the EC2 instance to Amazon RDS MySQL. This provides automated backups and independent management.

4. Backup or recovery configuration for the database.



This screenshot shows our RDS backup configuration. We have automated backups enabled with a 7-day retention period. This means if our database ever fails, we can restore data from the last 7 days, ensuring we don't lose critical student records

Rubric bands and score ranges

Band	Range	Descriptor
Unsatisfactory	0-3	Solution remains single-instance or single-subnet, with no meaningful high-availability design.
Needs Improvement	4-5	Multiple instances exist but the design still has a single-AZ or other major availability weakness.
Good	6-8	Web instances span two Availability Zones, the network is separated into public and private tiers, and managed services are used appropriately.

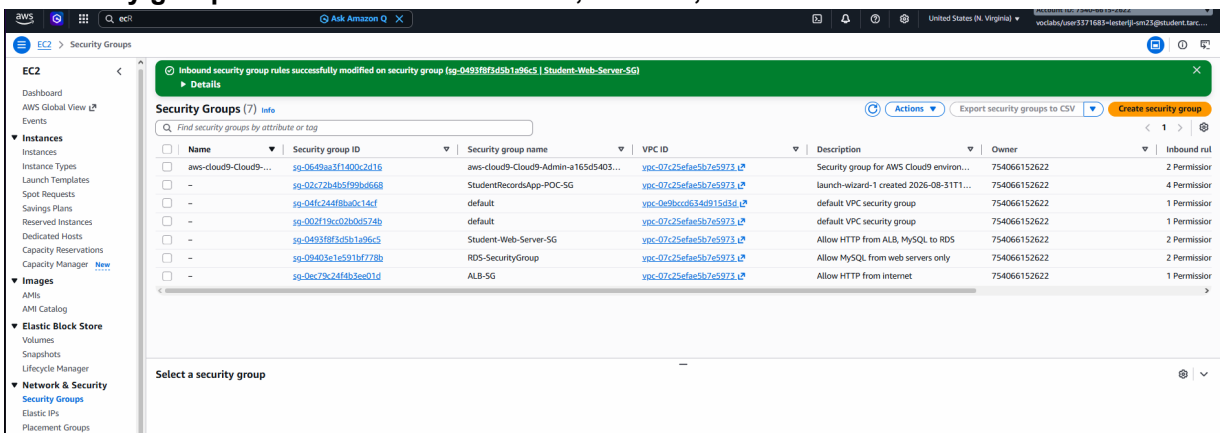
Excellent	9-10	The design meets the Good level and also shows recovery readiness, such as database backups and clear failover-oriented layout.

C5. Secure (10 marks)

The solution must apply core AWS security controls, especially around network access, credential handling, and private database placement.

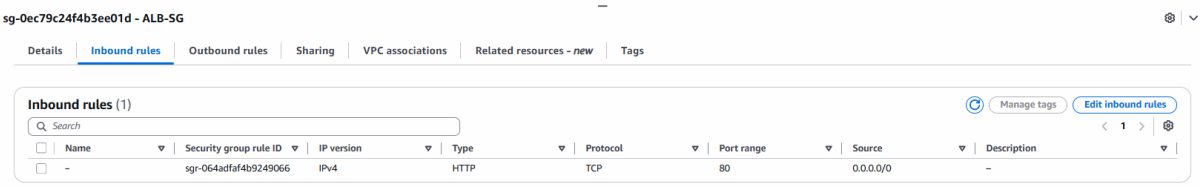
Required evidence in the report

1. Security group rules for the load balancer, web tier, and database tier.



Security Groups for the Application Load Balancer, Web Server, RDS Database

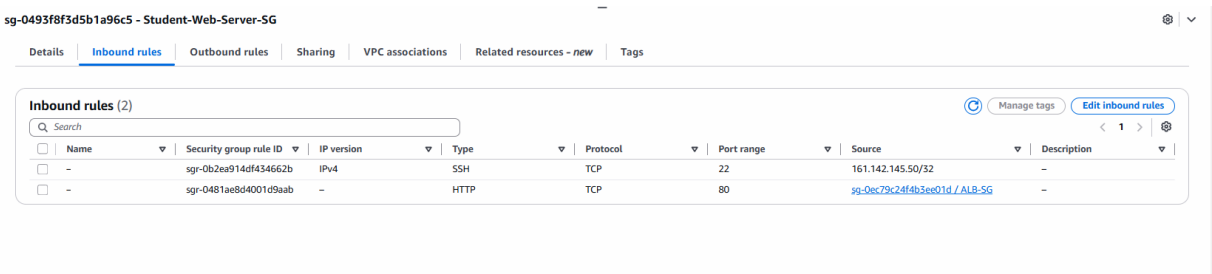
ALB Security Group Inbound Rule -



Application Load Balancer only has one inbound rule which allows HTTP from the whole internet

HTTP (80) from 0.0.0.0/0

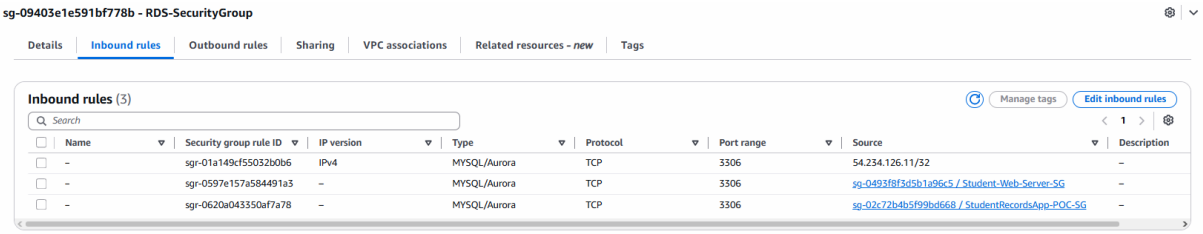
Web-Server Security Group Inbound Rule -



Web Server Security Group: Inbound – HTTP (80) from ALB-SG

Web-Security server is connected to The Application Load Balancer, to ensure it does not connect anywhere but only through the Application Load Balancer Security Group

RDS-SG Inbound Rule -



RDS Security Group: Inbound – MySQL (3306) from Web-Server-SG

This screenshot shows our security group chain. The ALB accepts HTTP from the internet. The web servers only accept traffic from the ALB security group – not directly from the internet. And RDS only accepts MySQL traffic from the web server security group. This ensures the database is never directly exposed to the internet and follows the principle of least privilege.

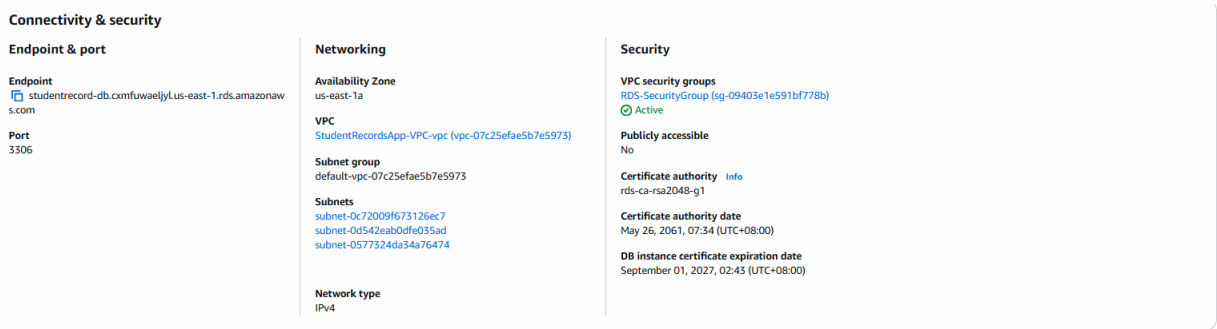
The private subnets have no direct internet access, ensuring the database is not exposed to the public internet. Security groups enforce strict access control:

The ALB accepts HTTP from the internet.

Web servers accept traffic only from the ALB.

RDS accepts MySQL traffic only from the web servers.

2. Evidence that the database is in a private subnet and not directly exposed to the internet.

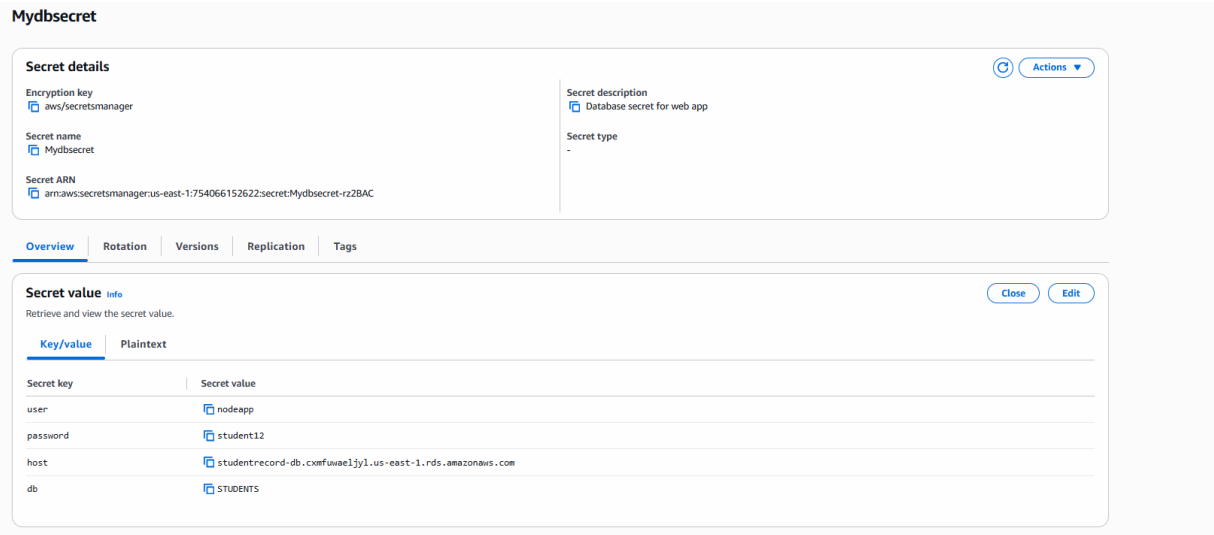


RDS configuration showing "Public accessibility: No"

The database is in a private subnet with no internet gateway

This screenshot shows our RDS database is not publicly accessible. The 'Public accessibility' is set to No, and it's placed in a private subnet with no internet gateway. This means nobody from the internet can directly access our database – only the web servers can communicate with it.

3. Proof that credentials are stored in AWS Secrets Manager rather than hardcoded in application code.

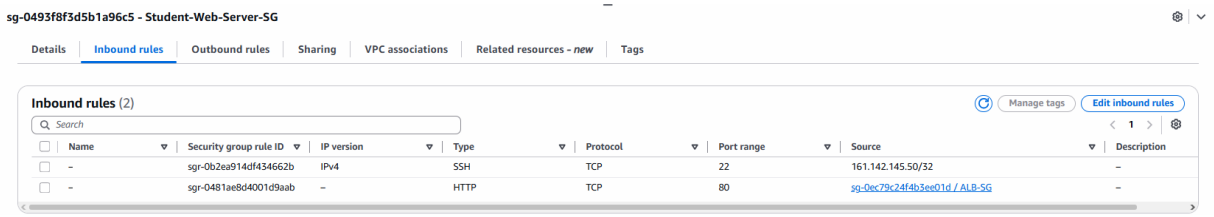


AWS Secrets Manager console showing the secret Mydbsecret

The secret contains: user, password, host, db

This screenshot shows our database credentials stored in AWS Secrets Manager. The secret contains the username, password, host, and database name. The application retrieves these credentials at runtime via the IAM role, so credentials are never hardcoded in the application code.

4. Evidence that administrative access is restricted to a controlled IP range or approved path.



Security group inbound rules for Web-Server-SG showing SSH (port 22) is restricted to the administrator's IP address (SEC-1). This ensures that only authorized users can access the EC2 instances for maintenance and no one can SSH into the servers

Rubric bands and score ranges

Band	Range	Descriptor
Unsatisfactory	0-3	Security controls are missing or major exposures remain, such as a public database or hardcoded credentials.
Needs Improvement	4-5	Some security controls exist, but important rules remain weak, overexposed, or only partially applied.
Good	6-8	The design secures most resources correctly, but there are

		still small gaps in least privilege, ingress restriction, or evidence quality.
Excellent	9-10	Security controls are consistently applied: least-privilege access is enforced, the database is private, credentials are externalized, and ingress rules are tightly scoped.

C6. Cost Optimized (10 marks)

The architecture must be sized deliberately and supported by a clear 12-month AWS Pricing Calculator estimate.

Required evidence in the report

1. AWS Pricing Calculator estimate for 12 months in us-east-1.

aws

Contact your AWS representative: Contact Sales

Export Date: 09/09/2026 **Language:** English

Estimate url

Estimate summary

Upfront cost	Monthly cost	Total 12 months cost
0.00 USD	84.00 USD	1,008.00 USD Includes upfront cost

Detailed Estimate

Name	Group	Region	Upfront cost	Monthly cost
Amazon EC2	-	US East (N. Virginia)	0.00 USD	12.08 USD
Status	-			
Description:	-			
Config summary	Tenancy (Shared Instances), Operating system (Ubuntu Pro), Workload (Consistent, Number of instances: 2), Advance EC2 instance (t3.micro), Pricing strategy (Compute Savings Plans 3yr No Upfront), Enable monitoring (disabled), EBS Storage amount (8 GB), DT Inbound: Not selected (0 TB per month), DT Outbound: Not selected (0 TB per month), DT Intra-Region: (0 TB per month)			

Name	Group	Region	Upfront cost	Monthly cost
Amazon RDS for MySQL	-	US East (N. Virginia)	0.00 USD	54.86 USD
Status	-			
Description:	-			
Config summary	Storage amount (20 GB), Storage for each RDS instance (General Purpose SSD (gp2)), Nodes (1), Instance type (db.t3.micro), Utilization (On-Demand only) (100 %Utilized/Month), Deployment option (Single-AZ), Pricing strategy (OnDemand)			

Name	Group	Region	Upfront cost	Monthly cost
Elastic Load Balancing	-	US East (N. Virginia)	0.00 USD	16.66 USD
Status	-			
Description:	-			
Config summary	Number of Application Load Balancers (1)			

Name	Group	Region	Upfront cost	Monthly cost
AWS Secrets Manager	-	US East (N. Virginia)	0.00 USD	0.40 USD
Status	-			
Description:	-			
Config summary	Number of secrets (1), Average duration of each secret (30 days), Number of API calls (1 per month)			

Acknowledgement

AWS Pricing Calculator provides only an estimate of your AWS fees and doesn't include any taxes that might apply. Your actual fees depend on a variety of factors, including your actual usage of AWS services. Learn more

Evidence ID: CO-1

- Amazon EC2 (2 × t3.micro, Compute Savings Plans 3yr No Upfront): Matches the Auto Scaling group's minimum/desired capacity of 2 instances. The 3-year Compute Savings Plan further reduces EC2 costs compared to On-Demand pricing.
- Amazon RDS (db.t3.micro, Single-AZ, 20 GB gp2): Single-AZ per the assignment's assumption, roughly halving RDS cost versus a Multi-AZ deployment. While the web tier spans two Availability Zones for high availability, the database is permitted to remain in a single AZ as per project requirements.
- Application Load Balancer: A single ALB distributes traffic across both Availability Zones, avoiding duplicate fixed hourly charges while ensuring high availability and load balancing for the web tier.
- AWS Secrets Manager: Only one secret is stored (database credentials), keeping this cost under USD 0.50/month.
- Data Transfer: Minimal data transfer costs as the application is a POC-level deployment with limited traffic. The estimate assumes conservative data transfer volumes.

The estimated cost of USD 1,008.00 over 12 months is proportionate to a POC-scale deployment and leaves headroom within a typical lab budget, while every instance family chosen (t3.micro / db.t3.micro) is the smallest burstable class capable of running this workload without out-of-memory errors or performance degradation.

2. Sizing rationale for EC2, Auto Scaling minimums, and Amazon RDS.

EC2 Instance Type (t3.micro):

- Why not larger? The application is a lightweight Node.js CRUD application. During load testing, t3.micro instances maintained adequate performance under normal and variable load (mean latency ~45-82ms). Under peak load, errors occurred due to database connection limits rather than CPU exhaustion, making larger instances unnecessary.

- Why not smaller (t2.nano)? t2.nano has only 0.5 GiB of memory, which would be insufficient for Node.js + npm dependencies + the operating system, causing out-of-memory errors

Auto Scaling Capacity:

- **Minimum:** 2 instances - This ensures high availability across the two availability zones (One instance per AZ)
- **Desired:** 2 instances - this maintains as a baseline capacity for normal traffic while keeping the costs minimal.
- **Maximum:** 4 instances - This allows for scale out during peak load without over provisioning.

Amazon RDS (db.t3.micro):

- Chosen for POC workload with Single-AZ deployment to reduce costs while meeting functional requirements. Although the web tier spans two Availability Zones for high availability, the project assumptions state: "The database is hosted only in a single Availability Zone," so this is a deliberate cost-saving decision that aligns with project requirements.
- 20GB gp2 storage provides a sufficient capacity for the students data.

3. Evidence that the architecture avoids unnecessary overprovisioning.

- **EC2 Instances:** Only 2 instances are deployed as a baseline, scaling up only when the CPU exceeds a 70% threshold.

- **Instance Type:** The t3.micro has been chosen on load testing results which proves the adequate performance.
- **RDS:** Single-AZ deployment, as project assumptions allow database in single Availability Zone which reduces the cost by 50%.
- **EBS Storage:** Only 8GB per EC2 instance and 20GB for RDS which is sufficient for the lightweight application.
- **Savings Plans:** Applied a 3 year compute savings plans (no upfront) to EC2 costs for optimal pricing.

4. Short note on budget trade-offs and cost-aware design decisions.
    - **Balance between performance and cost:** t3.micro instances provide the right balance for a lightweight Node.js application. Larger instances would increase cost without significant performance gains.
    - **High availability vs. cost:** Deployed minimum 2 instances for HA instead of a single instance, adding minimal cost ($12.08/month) for significant resilience.
    - **Managed services:** Using RDS and ALB increases cost compared to self-managed alternatives but reduces operational overhead and improves reliability.
    - **Savings Plans:** EC2 costs reduced by committing to 3-year Compute Savings Plans, lowering monthly costs.
    - **Total monthly cost of $84.00** keeps the solution within the lab budget constraints while meeting all functional and non-functional requirements.

Rubric bands and score ranges

Band	Range	Descriptor
Unsatisfactory	0-3	Cost was not considered and resource sizes appear arbitrary or clearly excessive.
Needs Improvement	4-5	A cost estimate or sizing plan exists, but one or more components remain obviously over- or under-sized.
Good	6-8	Most components are sized appropriately and the design accounts for baseline traffic without obvious waste.
Excellent	9-10	Resource sizing is deliberate, the calculator estimate is complete, and the design balances resilience, performance, and budget effectively.

C7. High Performing (10 marks)

The application must remain responsive under normal, variable, and peak load without timeouts or unacceptable response degradation.

Required evidence in the report

1. Load test method, tool, and test duration.

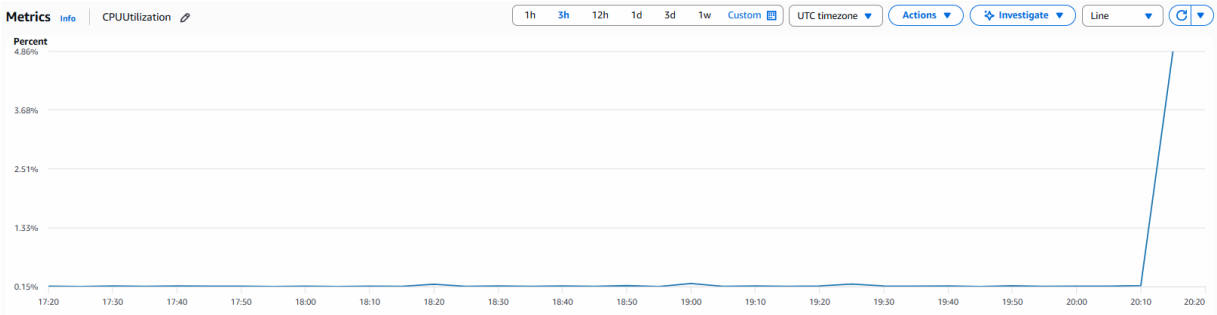
```
Target URL:          http://studentrecord-alb-411902315.us-east-1.elb.amazonaws.com/students
Max time (s):        60
Target rps:          300
Concurrent clients:  865
Agent:               keepalive

Completed requests:  17891
Total errors:        16
Total time:          60.001 s
Mean latency:        753.4 ms
Effective rps:       298

Percentage of requests served within a certain time
  50%      20 ms
  90%      2363 ms
  95%      3278 ms
  99%      10003 ms
 100%      20503 ms (longest request)
```

2. Results for normal, variable, and peak load conditions.

Normal

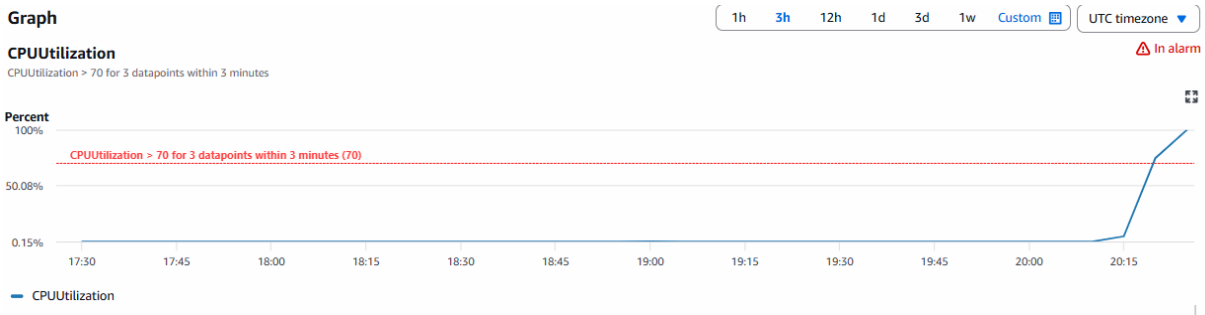


```
Target URL:          http://studentrecord-alb-411902315.us-east-1.elb.amazonaws.com/students
Max time (s):        60
Target rps:          1000
Concurrent clients:  4182
Agent:               keepalive

Completed requests:  56021
Total errors:        5940
Total time:          60.001 s
Mean latency:        2023.4 ms
Effective rps:       934

Percentage of requests served within a certain time
  50%      237 ms
  90%      7279 ms
  95%      10055 ms
  99%      28662 ms
 100%      30025 ms (longest request)
```

Peak



During the peak load test (1000 rps, 500 concurrent clients), the CloudWatch CPU utilization metrics show a clear spike above the 70% target tracking threshold, triggering the Auto Scaling group to scale out from 2 to 4 instances

3. Response-time summary and whether errors or timeouts occurred.

Load Level	Completed Requests	Total Errors	Mean Latency	Effective RPS
Normal	17,891	16	753.4 ms	298
Peak	56,021	5940	2023.4 ms	934

4. Screenshots or metrics from CloudWatch, the load test tool, or both.

Rubric bands and score ranges

Band	Range	Descriptor
Unsatisfactory	0-3	Application struggles even at normal load or fails significantly under increased load.
Needs Improvement	4-5	Application works at normal load but times out, degrades sharply, or becomes unstable during higher load.
Good	6-8	Application remains functional under normal and higher load, but response time worsens noticeably at peak load.
Excellent	9-10	Application remains responsive and stable under normal, variable, and peak load, with routine operations finishing within the expected time.

Conclusion

The project successfully deployed a highly available, scalable web application on AWS for Example University's student records system, achieving all functional and non-functional requirements. The architecture leverages Application Load Balancer, Auto Scaling, multi-AZ deployment, Amazon RDS, and AWS Secrets Manager to deliver a resilient, secure, and cost-optimized solution.

All CRUD operations perform without perceivable delay, load balancing distributes traffic evenly across two Availability Zones, and Auto Scaling responds effectively to demand. Security controls are consistently enforced, and the estimated cost of $84/month ($1,008/year) remains within budget.

This project demonstrates the successful application of AWS Well-Architected Framework best practices, providing a solid foundation for future production enhancements.

D. Demonstration Session and Student Effort (30 marks per student)

This section is individual, not shared. Each student should be able to demonstrate, explain, and defend their own contribution during the session.

ID	Criterion	Max	What full marks look like
D1	Demo presentation	10	Presents the deployed solution clearly, demonstrates the key functions, and answers questions about the architecture and implementation.
D2	Tutorial and practical submission	10	Shows complete and properly organized tutorial and practical submission evidence that supports the final assignment outcome.
D3	AWS guided lab and challenge lab	10	Shows credible completion evidence for the relevant AWS guided lab and challenge lab work that supports the assignment build.

D3. AWS guided lab and challenge lab

Evidence of **Natalie Koa Hao Yee** guided lab and challenge lab completed:

Grades for nataliekhy-sm23@student.tarc.edu.my

Print Grades (javascript:window.print())

Course: AWS Academy Cloud Archi

Arrange By: Due Date

Apply

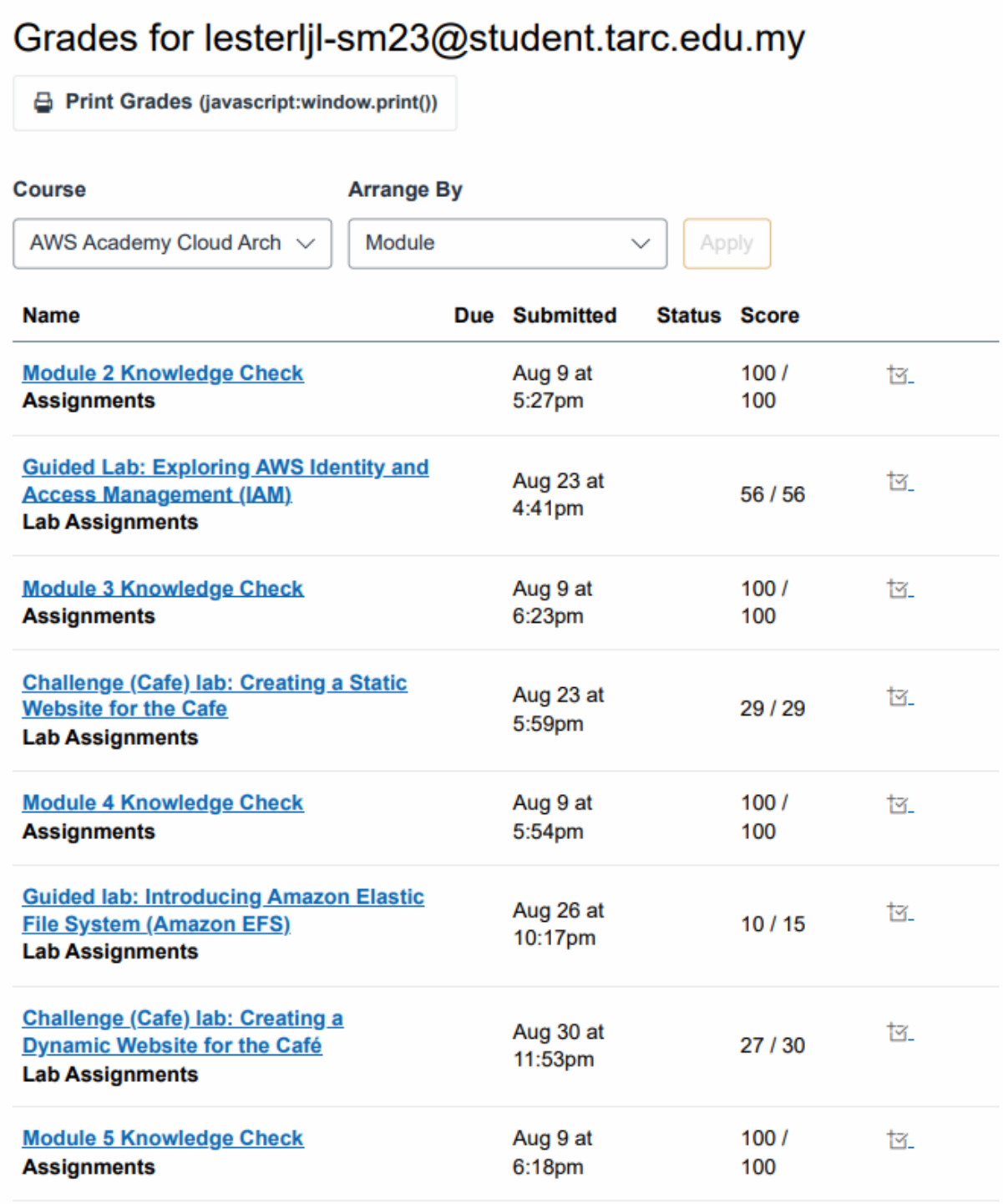
Name	Due	Submitted	Status	Score
Challenge (Cafe) lab: Creating a Dynamic Website for the Café Lab Assignments		Aug 26 at 11:54pm		30 / 30
(Optional) Guided lab: Breaking a Monolithic Node.js Application into Microservices Lab Assignments				- / 56
Academy Cloud Architecting Course Assessment Assignments				- / 100
Capstone Project Lab Assignments				- / 35
Capstone Project Lab Assignments				- / 56
Challenge (Café) lab: Automating Infrastructure Deployment Lab Assignments		Aug 30 at 9:19am		56 / 56
Challenge (Café) lab: Creating a Scalable and Highly Available Environment for the Café Lab Assignments		Aug 30 at 7:54am		56 / 56

Name	Due	Submitted	Status	Score	
Challenge (Cafe) lab: Creating a Static Website for the Cafe **Lab Assignments**		Aug 30 at 3:34am		29 / 29	
Challenge (Cafe) lab: Creating a VPC Networking Environment for the Café **Lab Assignments**		Aug 30 at 5:55am		56 / 56	
Challenge (Café) lab: Implementing a Serverless Architecture for the Café **Lab Assignments**		Aug 30 at 10:28am		56 / 56	
Challenge (Cafe) lab: Migrating a Database to Amazon RDS **Lab Assignments**		Aug 30 at 4:17am		25 / 25	
Guided lab: Automating Infrastructure with AWS CloudFormation **Lab Assignments**		Aug 27 at 11:46pm		56 / 56	
Guided lab: Building Decoupled Applications by Using Amazon SQS **Lab Assignments**		Aug 28 at 4:43am		56 / 56	
Guided Lab: Configuring Hybrid Storage and Migrating Data with AWS Storage Gateway S3 File Gateway **Lab Assignments**		Aug 28 at 8:42pm		40 / 40	
Guided lab: Creating a Highly Available Environment **Lab Assignments**		Aug 30 at 6:35am		56 / 56	
Guided lab: Creating a Virtual Private Cloud **Lab Assignments**		Aug 27 at 2:08am		56 / 56	
Guided lab: Creating a VPC Peering Connection **Lab Assignments**		Aug 27 at 3:37am		56 / 56	

Name	Due	Submitted	Status	Score	
Guided lab: Creating an Amazon RDS Database Lab Assignments		Aug 27 at 10:41am		20 / 20	
Guided lab: Encrypting Data at Rest by Using AWS Encryption Options Lab Assignments		Aug 27 at 2:54am		56 / 56	
Guided Lab: Exploring AWS Identity and Access Management (IAM) Lab Assignments		Aug 15 at 10:47pm		56 / 56	
Guided lab: Implementing a Serverless Architecture on AWS Lab Assignments		Aug 28 at 7:21pm		56 / 56	
Guided lab: Introducing Amazon Elastic File System (Amazon EFS) Lab Assignments		Aug 26 at 9:49pm		15 / 15	
Guided lab: Securing Applications by using Amazon Cognito Lab Assignments		Aug 27 at 8:10pm		56 / 56	
Module 2 Knowledge Check Assignments		Sep 6 at 9:38pm		100 / 100	
Module 3 Knowledge Check Assignments		Aug 10 at 10:27am		100 / 100	
Module 4 Knowledge Check Assignments		Aug 10 at 10:30am		100 / 100	
Module 5 Knowledge Check Assignments		Aug 10 at 10:34am		100 / 100	
Module 6 Knowledge Check Assignments		Aug 10 at 10:41am		100 / 100	

Name	Due	Submitted	Status	Score	
Module 7 Knowledge Check Assignments		Aug 10 at 10:46am		100 / 100	
Module 8 Knowledge Check Assignments		Aug 15 at 9:23pm		100 / 100	
Module 9 Knowledge Check Assignments		Sep 8 at 6:56am		100 / 100	
Module 10 Knowledge Check Assignments		Aug 15 at 11:09pm		100 / 100	
Module 11 Knowledge Check Assignments		Aug 15 at 11:13pm		100 / 100	
Module 12 Knowledge Check Assignments		Aug 15 at 11:18pm		100 / 100	
Module 13 Knowledge Check Assignments		Aug 15 at 11:22pm		100 / 100	
Module 14 Knowledge Check Assignments		Aug 16 at 1:29am		100 / 100	
Module 15 Knowledge Check Assignments		Aug 16 at 1:40am		100 / 100	
Module 16 Knowledge Check Assignments		Sep 8 at 7:11am		100 / 100	
Lab Assignments				100%	887.00 / 887.00
Assignments				100%	1,500.00 / 1,500.00
Total				100%	2,387.00 / 2,387.00

Evidence of **Lester Liau Jiun Lang** guided lab and challenge lab completed:



Grades for lesterljl-sm23@student.tarc.edu.my

Print Grades (javascript:window.print())

Course: AWS Academy Cloud Arch

Arrange By: Module

Apply

Name	Due	Submitted	Status	Score
Module 2 Knowledge Check **Assignments**		Aug 9 at 5:27pm		100 / 100
Guided Lab: Exploring AWS Identity and Access Management (IAM) **Lab Assignments**		Aug 23 at 4:41pm		56 / 56
Module 3 Knowledge Check **Assignments**		Aug 9 at 6:23pm		100 / 100
Challenge (Cafe) lab: Creating a Static Website for the Cafe **Lab Assignments**		Aug 23 at 5:59pm		29 / 29
Module 4 Knowledge Check **Assignments**		Aug 9 at 5:54pm		100 / 100
Guided lab: Introducing Amazon Elastic File System (Amazon EFS) **Lab Assignments**		Aug 26 at 10:17pm		10 / 15
Challenge (Cafe) lab: Creating a Dynamic Website for the Café **Lab Assignments**		Aug 30 at 11:53pm		27 / 30
Module 5 Knowledge Check **Assignments**		Aug 9 at 6:18pm		100 / 100

Name	Due	Submitted	Status	Score	
Guided lab: Creating an Amazon RDS Database **Lab Assignments**		Aug 26 at 10:36pm		20 / 20	
Challenge (Cafe) lab: Migrating a Database to Amazon RDS **Lab Assignments**		Sep 11 at 5:48am		23 / 25	
Module 6 Knowledge Check **Assignments**		Aug 9 at 6:40pm		100 / 100	
Guided lab: Creating a Virtual Private Cloud **Lab Assignments**		Aug 27 at 1:55am		56 / 56	
Challenge (Cafe) lab: Creating a VPC Networking Environment for the Café **Lab Assignments**		Sep 11 at 3:02pm		55 / 56	
Module 7 Knowledge Check **Assignments**		Aug 9 at 7:20pm		100 / 100	
Guided lab: Creating a VPC Peering Connection **Lab Assignments**		Aug 27 at 12:24am		28 / 56	
Module 8 Knowledge Check **Assignments**		Aug 16 at 2:57pm		100 / 100	
Guided lab: Securing Applications by using Amazon Cognito **Lab Assignments**		Aug 30 at 8:32pm		9.33 / 56	
Guided lab: Encrypting Data at Rest by Using AWS Encryption Options **Lab Assignments**		Aug 27 at 6:52pm		56 / 56	
Module 9 Knowledge Check **Assignments**		Aug 16 at 11:25pm		100 / 100	

Name	Due	Submitted	Status	Score	
Guided lab: Creating a Highly Available Environment **Lab Assignments**		Aug 28 at 2:29am		52.27 / 56	
Challenge (Café) lab: Creating a Scalable and Highly Available Environment for the Café **Lab Assignments**		Sep 11 at 4:52pm		54.19 / 56	
Module 10 Knowledge Check **Assignments**		Aug 16 at 11:30pm		100 / 100	
Guided lab: Automating Infrastructure with AWS CloudFormation **Lab Assignments**		Aug 27 at 2:33am		56 / 56	
Challenge (Café) lab: Automating Infrastructure Deployment **Lab Assignments**		Sep 11 at 4:06pm		52.27 / 56	
Module 11 Knowledge Check **Assignments**		Aug 16 at 11:54pm		100 / 100	
Module 12 Knowledge Check **Assignments**		Aug 17 at 10:43am		100 / 100	
Guided lab: Building Decoupled Applications by Using Amazon SQS **Lab Assignments**		Aug 30 at 9:20pm		37.33 / 56	
Module 13 Knowledge Check **Assignments**		Aug 19 at 12:18am		100 / 100	
Guided lab: Implementing a Serverless Architecture on AWS **Lab Assignments**		Aug 28 at 4:34pm		56 / 56	
(Optional) Guided lab: Breaking a Monolithic Node.js Application into Microservices				- / 56	

Name	Due	Submitted	Status	Score	
Lab Assignments					
Challenge (Café) lab: Implementing a Serverless Architecture for the Café **Lab Assignments**		Sep 11 at 5:43pm		48 / 56	
Module 14 Knowledge Check **Assignments**		Aug 19 at 12:25am		100 / 100	
Module 15 Knowledge Check **Assignments**		Aug 19 at 1am		100 / 100	
Guided Lab: Configuring Hybrid Storage and Migrating Data with AWS Storage Gateway S3 File Gateway **Lab Assignments**		Sep 11 at 6:48pm		5 / 40	
Module 16 Knowledge Check **Assignments**		Aug 19 at 1:07am		100 / 100	
Capstone Project **Lab Assignments**				- / 35	
Academy Cloud Architecting Course Assessment **Assignments**		Aug 30 at 9:33pm		84 / 100	
Capstone Project **Lab Assignments**				- / 56	
Lab Assignments				82.34%	730.39 / 887.00
Assignments				99%	1,584.00 / 1,600.00
Total				93.06%	2,314.39 / 2,487.00

References

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https://docs.aws.amazon.com/autoscaling/ec2/userguide/[reference:6]

Appendices

Appendix A: Group Contact Form

Group contact form at the front of this report for the assigned roles, tasks, and signature.

Appendix B: Google Slides Presentation

The complete presentation slides for this project are available at:

https://docs.google.com/presentation/d/1iVymZyF_J9QeKY55jtm0PPztbsjg7FEi/edit?usp=sharing&ouid=104745351632863241870&rtpof=true&sd=true

The slides include:

- explanation of C1 to C7 in slide format.

Appendix C: AWS Pricing Calculator Estimate

The full 12-month cost estimate for the solution in us-east-1 is available at the following AWS Pricing Calculator link:

https://calculator.aws/#/estimate